



### Matrix Partners raising \$450m

Matrix Partners are raising \$450m for new India fund, doubling down on South Asian market. <https://digit.in/jul22-10>



### Self Plugging Eye Microneedle

Scientists have discovered a self-plugging eye microneedle for better drug application to the human eye. <https://digit.in/jul22-11>

development stage. The prototype will be available by the end of this year, with mass production beginning in the second half of 2024. The AR Glass is going to be based on this waveguide technology. Unlike a mixed reality AR/VR machine that Apple plans to introduce later this year, the AR Glass would be entirely focused on augmented reality technologies. According to a study, AR Glass could eventually surpass the iPhone in ten years. Additionally, an innovative periscope lens is expected to be included in the iPhone 15 Pro, however it may only be for the Max variant and not the Pro. A periscope lens allows for optical zooming not achievable with regular smartphone cameras, allowing for 5x or even 10x optical zoom. 2022 to 2024 are Hot years for Apple releases, and we can't wait for what's in store!



### NASA'S Z-1 SPACESUIT

NASA's Z-1 spacesuit prototype, which was revealed a decade ago, is eerily similar

to the otherworldly gear worn by the famed space ranger in Toy Story. While the project is no longer active, contractors can use the development history to design their own spacesuits for future agency needs, such as moon expeditions. In 2012, the green-striped Z-1 suit was put to the test to see how well futuristic spacesuits might cope with a "softer" appearance. With the recent release of the movie "Light-year", the suit prototype is garnering a lot of attention. The Z-1 was a testbed spacesuit designed to improve astronaut mobility during surface spacewalks by allowing them to bend and twist more easily than they could in Apollo-era spacesuits in the 1960s and 1970s. Increased mobility was achieved utilising redesigned bearings in the shoulder and hip joints, allowing space suit users to dip, walk, and bend with ease, all critical actions for a planetary explorer gathering samples or navigating through rugged terrain. The upper torso was more softer than previous spacesuit versions, and astronauts could enter the suit

via a "spaceport," which is a design that enabled an astronaut to enter the spacesuit from a vehicle. The spacesuit remains docked outside the ship, reducing storage and dust mitigation requirements. The Z-1 was never meant to get into space. However, it was improved upon over the next two years, and a new model was built. Since the prototype is still in commission, there is a possibility for some Hot developments to feature a functioning Z-1 suit in a few years.



### DRM IN PAPER

The printing industry has deeper politics than you thought. Basically everything that goes wrong in other industries. Printer makers have used copyright laws to target refillers, sent out phoney "security upgrades" to fool you into installing code that blocks third-party ink, deceived and lied to prevent "security chips" from being extracted from e-waste and used in new cartridges, and more. However, one thing that printer companies cannot exploit is digital media rights, because they deal with physical printing paper, right? Wrong. Dymo, a popular company that provides products for label-making, has managed to put DRM (Digital Rights Management) onto physical paper. Of course, printing and label paper is thin and clean, and there is no

way to insert a chip to enforce DRM. But what about the rolls in which the labels are sold? Dymo just announced two new versions of their label-makers, the 550 and 5XL, with DRM that bans the use of third-party labels. The new label rolls include a booby-trap: an RFID-enabled microcontroller that verifies your label maker, proving that you purchased Dymo's heavily-priced labels rather than those from competitors. The chip counts down the labels as you print them (thus it won't work with a standard label roll). The gesture in itself is somewhat unethical, but it is made worse because Dymo has not made the use of this new feature transparent. While smart Dymo users are shifting to cheaper alternatives, we do Not appreciate this kind of deceit.



### TELEGRAM PREMIUM

Telegram, which has over 700 million monthly active users, is launching a premium tier with additional capabilities as part of its effort to monetize a piece of its vast user base. The monthly subscription appears to cost between \$4.99 and \$6. The Premium tier allows users to send files up to 4GB (up from 2GB) and offers speedier downloads, according to Telegram. Customers will be able to follow up to 1,000 channels (up from 500 for free users), as well as establish up to 20 conversation folders with up to 200 talks each. Telegram Premium users will be able to add up to four identities and pin up to ten chats to their profile. Telegram's founder and CEO Pavel Durov stated earlier this month that the decision to establish a premium tier was made in response to user demand for more storage and bandwidth. Keeping user first, One new free feature is where all users can join a public group without an invite link by submitting a join request. Verified groups and channels will be able to display their emblem at the top of the chat, which is another new feature targeted at free users. Ultimately, there is value to both the paid and free versions of Telegram, which is very Hot. **H**